$$\tau(n) = \sum_{g^2 \mid n} \chi_5(g) g^{10} R(n/g^2)$$

Möbius inversion on square divisors

$$R(n) = \sum_{d^2 \mid n} \mu(d) \chi_5(d) d^{10} \tau(n/d^2)$$

$R(p) = \tau(p)$ at every prime. Extraction preserves the prime target.